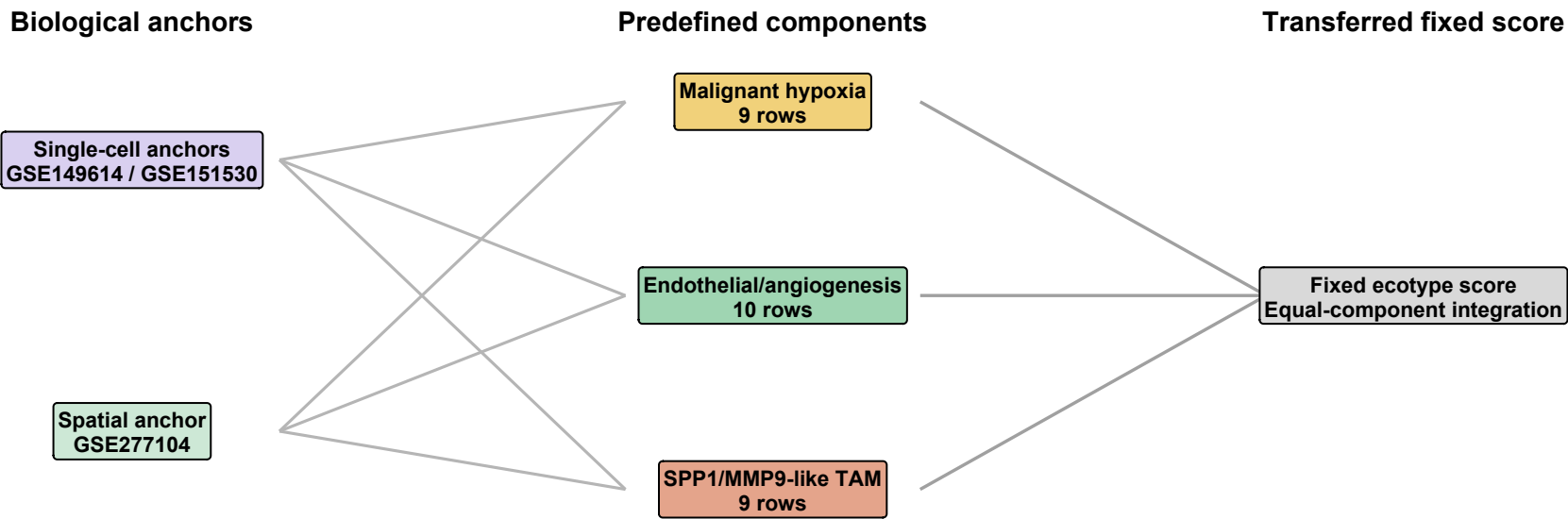
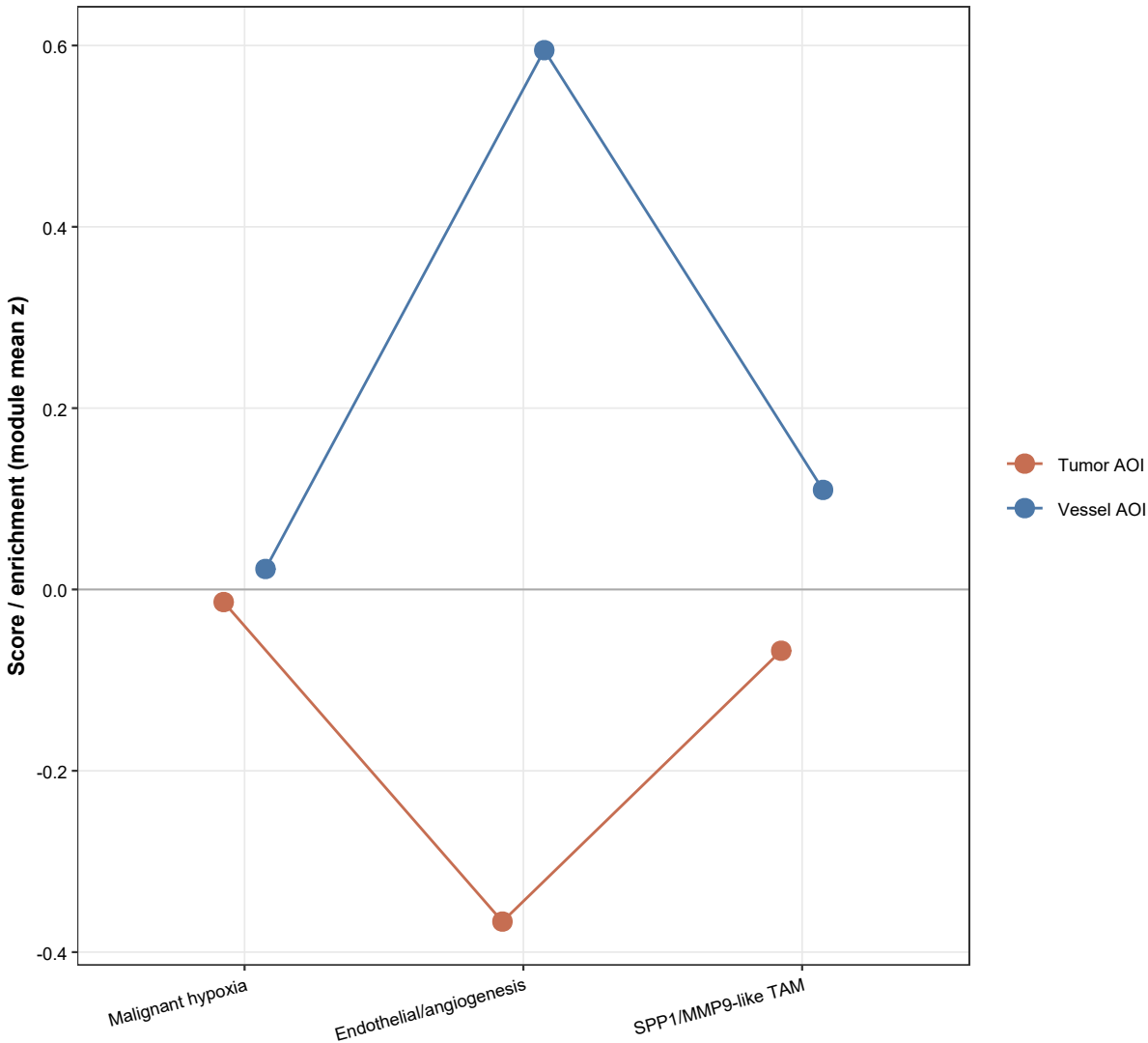


A Anchored construction of the hypoxia-adaptive tumor-endothelial-macrophage ecotype
Single-cell and spatial support define a fixed three-component ecotype transferred without response-guided revision



B Spatial anchor summary
Module-level mean z-summary from Tumor and Vessel AOIs (GSE277104)



C Gene-level support and transferability matrix
VEGFA is duplicated across malignant hypoxia and TAM support; DLL4 is excluded before downstream transfer

ANGPT2	0.62	0.49	Yes	Unique	Retained	Endothelial/angiogenesis
CD34	0.90	0.81	Yes	Unique	Retained	
CXCL12	0.25	0.37	Yes	Unique	Retained	
DLL4	NA	0.71	No	Unique	Excluded	
ENG	1.24	0.81	Yes	Unique	Retained	
FLT1	1.11	0.62	Yes	Unique	Retained	
JAG1	0.34	0.58	Yes	Unique	Retained	
KDR	0.63	0.75	Yes	Unique	Retained	
LGALS9	0.17	0.50	Yes	Unique	Retained	
NOTCH1	0.27	0.76	Yes	Unique	Retained	
PECAM1	1.26	0.81	Yes	Unique	Retained	
VEGFA	0.40	0.37	Yes	Dup	Retained	Malignant hypoxia
ADM	0.22	-0.27	Yes	Unique	Retained	
ALDOA	0.56	0.05	Yes	Unique	Retained	
BNIP3	1.24	-0.02	Yes	Unique	Retained	
DDIT4	-0.07	-0.13	Yes	Unique	Retained	
ENO1	0.98	-0.07	Yes	Unique	Retained	
HK2	0.03	-0.11	Yes	Unique	Retained	
PDK1	0.12	0.06	Yes	Unique	Retained	
SLC2A1	0.10	-0.01	Yes	Unique	Retained	
APOE	1.08	NA	Yes	Unique	Retained	SPP1/MMP9-like TAM
C1QA	2.51	NA	Yes	Unique	Retained	
C1QB	2.58	NA	Yes	Unique	Retained	
CD68	1.56	NA	Yes	Unique	Retained	
LGALS3	0.87	NA	Yes	Unique	Retained	
MARCO	0.61	NA	Yes	Unique	Retained	
MMP9	0.56	NA	Yes	Unique	Retained	
SPP1	1.28	NA	Yes	Unique	Retained	
VEGFA	0.14	NA	Yes	Dup	Retained	
Support / indicator						
0.0 0.5 1.0						